



Gamaliel Mendoza-Cuevas

Data Engineer building scalable ETL pipelines, medallion data architectures, and analytics-ready data platforms on Databricks and Spark.



Executive Summary

Professional Overview

A Data Engineer with a PhD in Neuroscience and hands-on ownership of enterprise data platforms. I design and operate ETL pipelines, tiered medallion (Bronze/Silver/Gold) architectures, and data quality frameworks that make large healthcare datasets reliable, governed, and analytics-ready.

I work across Databricks, Spark, SQL, and Python to deliver a single source of truth that accelerates analytics and machine learning organization-wide. I partner with analysts, data scientists, and senior leadership to turn raw, unstructured data into well-modeled datasets, communicating clearly at every organizational level.

Core Competencies & Professional Certifications 🖱️🔗📄

Data Engineering

Data Engineering · ETL Pipelines · Data Warehousing · Data Architecture · Medallion Architecture (Bronze/Silver/Gold) · Data Modeling · Data Quality · Data Integrity · Data Governance · Batch Processing · Single Source of Truth

Platforms & Databases

Databricks · Spark · Hadoop · Oracle · MySQL · SQL · MLflow · Cloud & On-Premise Environments

Languages & Tools

SQL · Python · Spark · R · MATLAB · PyTorch · TensorFlow · Scikit-learn · Git · GitHub · Bitbucket

Analytics & ML

Data Analytics · Statistical Modeling · Machine Learning · Deep Learning · Model Deployment · Model Serving · Feature Pipelines · Data Visualization

Professional / Soft Skills

Remote Work · Cross-functional Collaboration · Executive Reporting · Data Visualization · Stakeholder Communication · Bilingual (EN/ES) · German (A2)

Domain

Healthcare Analytics · Healthcare Data · Neuroscience · Translational Research · Signal Processing

Professional Experience

Cotiviti

Aug 2025 – Aug 2026 | Remote, Mexico City, Mexico.

Associate Data Scientist — Data Engineering & Platform Architecture.

- Built and scaled robust ETL pipelines across local and cloud (Databricks) environments, ensuring enterprise-grade data integrity and availability.
- Delivered a tiered Bronze/Silver/Gold (medallion) data architecture, establishing a single source of truth to accelerate organization-wide analytics.
- Implemented proactive data quality monitoring frameworks (MLflow), safeguarding data integrity, pipeline reliability, and regulatory compliance.
- Developed and deployed Python-based, ML-powered applications, translating advanced analytics into scalable, business-ready healthcare solutions.
- Deployed and maintained model-serving endpoints for neural network classifiers, enabling real-time, scalable healthcare data intelligence.
- Architected and operationalized end-to-end recommendation systems, driving measurable improvements in healthcare service delivery outcomes.
- Engineered K-means clustering-based recommendation engines to optimize resource allocation and elevate stakeholder value.
- Delivered high-performance CNN-based classification solutions, enhancing predictive accuracy across mission-critical healthcare data assets.
- Designed and implemented custom Byte-Pair Encoding (BPE) tokenization pipelines, optimizing downstream NLP performance on enterprise healthcare datasets.
- Led the fine-tuning of Transformer-based large language models, unlocking actionable insights from complex, unstructured healthcare data.

BIOINVERT

Jan 2017 – Jun 2017 | Mexico City, Mexico

Data Analyst.

Standardized and structured imaging datasets, building reproducible processing workflows for neuro- and physiological metrics.

- Conducted advanced CT imaging analysis to quantify bone density, supporting evidence-based clinical decision-making.
- Delivered statistical modeling of age-related bone density trends, generating actionable insights for longitudinal health research.

- Co-authored a peer-reviewed publication accepted in Veterinary Radiology and Ultrasound, reinforcing thought leadership in the field.

APREXBIO

Jan 2016 – Jan 2017 | Mexico City, Mexico

Research Intern.

Led data standardization protocols and structured acquisition of neurophysiological benchmark datasets.

- Executed analysis of visual evoked potentials.
- Achieved certification in MRI equipment operation and acquisition/analysis of anatomical imaging data.
- Co-authored a book chapter establishing standardized benchmark values for visual evoked potentials.
- Co-authored a peer-reviewed publication accepted in Experimental Gerontology.

Academic Credentials

PhD

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2020 – Apr 2026 | Queretaro, Mexico

Neuroscience.

Doctoral research portfolio spanning large-scale experimental data acquisition and processing, behavioral analytics, neurophysiology, and translational neuroscience, delivered within a top-tier Biomedical Sciences program.

- Delivered a peer-reviewed publication accepted in the Journal of Neurophysiology.
- Doctoral Thesis: Engineered a low-dose L-DOPA intervention strategy that measurably enhanced visual discrimination performance in a Parkinson's Disease model.
- Doctoral Advisor: Dr. Luis Alberto Carrillo Reid.
- Awarded a competitive doctoral scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation in recognition of research excellence.

Master

Institute of Neurobiology | National Autonomous University of Mexico | Aug 2018 – May 2020 | Queretaro, Mexico

Neurobiology.

Structured, processed, and analyzed behavioral datasets measuring visual discrimination performance within a translational Parkinson's Disease model.

- Graduated with Honorific Mention and a 9.16/10.00 GPA, placing in the top tier of the program cohort.
- Awarded a competitive Master's scholarship by the Mexican Secretariat of Science, Humanities, Technology, and Innovation.

Language Proficiencies

English

C1 — Professional Working Proficiency (TOEFL iBT: 110/120).

German

A2 — Elementary Proficiency, actively progressing (A1 certified via Goethe-Zertifikat).

Spanish

Native / Bilingual Proficiency.

Selected Publications

Impaired visually guided behavior in a mouse model of Parkinson's disease

Mendoza-Cuevas, G., Perez-Becerra, J., Velazquez-Contreras, R., Calderon, V., Carrillo-Reid, L.
Journal of Neurophysiology, 2025 | DOI: [10.1152/jn.00307.2025](https://doi.org/10.1152/jn.00307.2025)

Visual evoked potentials in the Rhesus monkey

Mendoza-Cuevas, G., Hernández-Godínez, B., Ibáñez-Contreras, A.
Book chapter, 2017 | ISBN: 9781536110753

Computed tomography is a feasible method for quantifying bone density in Macaca mulatta

Solís-Chávez, S., Castillo-Rivera, M., Arteaga-Silva, M., Ibáñez-Contreras, A., Hernández-Godínez, B., Morón-Mendoza, A.,
Mendoza-Cuevas, G., Morales-Guadarrama, A., Sacristan-Rock, E.
Veterinary Radiology and Ultrasound, 2018 | DOI: [10.1111/vru.12624](https://doi.org/10.1111/vru.12624)

Electrical activity of sensory pathways in female and male geriatric Rhesus monkeys (Macaca mulatta), and its relation to oxidative stress

Ibáñez-Contreras, A., Hernández-Arciga, U., Poblano, A., Arteaga-Silva, M., Hernández-Godínez, B., Mendoza-Cuevas, G., Toledo-Pérez, R., Alarcón-Aguilar, A., González-Puertos, V., Königsberg, M.
Experimental Gerontology, 2018 | DOI: [10.1016/j.exger.2017.11.003](https://doi.org/10.1016/j.exger.2017.11.003)

The effect of oxidative stress on brain electrical activity and its repercussions on sensory organization in geriatric Rhesus monkeys in captivity

Ibáñez-Contreras, A., Hernández-Godínez, B., Mendoza-Cuevas, G., Hernández-Arciga, U., Königsberg, M.
Book chapter, 2017 | ISBN: 9781536110753

Awards & Recognition

Second Place — 9th State Health Research Forum

Recognized for excellence in visual discrimination research within a Parkinson's Disease model • Querétaro, Mexico • 2024

Third Place — Latin American Brain Initiative (LATBrain) Design Competition

2021